

Multiple comparisons of means for synthetic fiber data set

In Lecture Notes no 1 we have seen that the null hypothesis

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

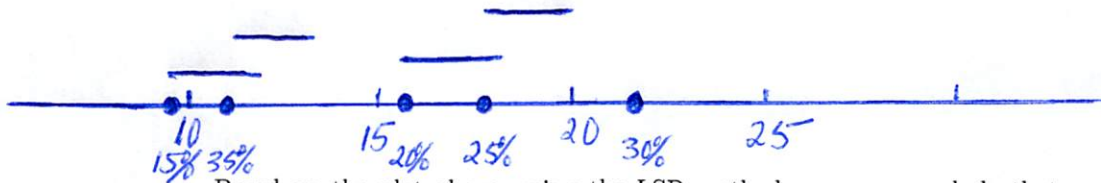
has been rejected. This implies that the average breaking strength for the five synthetic cotton blended fibers are not equal to each other. We now proceed with a multiple comparison of means to decide which of the synthetic cotton blends are the strongest. We will use the SAS software to implement this. First let us discuss how one can implement the multiple comparison of means with a calculator and the tables for percentage points of distributions. First let us use the LSD method.

In our data set for synthetic fiber, $N = 25$ and $n = 5$. The $MSE = 161.2/20 = 8.06$ and if we want the multiple testing via the LSD method to have significance level $\alpha = .05$, then we have to use in the formula for the LSD, $t_{.025;20} = 2.086$ which we get from Table II in the Appendix, page 700 in our textbook. Therefore, the LSD is equal to:

$$LSD = t_{.025;20} \sqrt{\frac{2MSE}{5}} = 2.086 \sqrt{\frac{2 \times 8.06}{5}} = 3.7455.$$

Based on this value of the LSD, we will declare any pair of population means to be significant different if their sample means differ by more than 3.7455.

Let us arrange the five sample means from smallest to largest as follows. We start from left to right and draw lines of length 3.7455 from the value of each sample mean



Based on the plot above, using the LSD method, one can conclude that the average (population mean) strength of synthetic fiber with 15% cotton and 30% cotton are not different from each other. The average (population mean) strength of synthetic fiber with 20% cotton and 25% cotton are not different from each other. The average strength of the synthetic fiber with 30% cotton is significantly higher than the rest.

The problem with the LSD method that actual significance level is much higher than the stated .05 level. This means that we can be wrong with our conclusion more than 5% of the time.

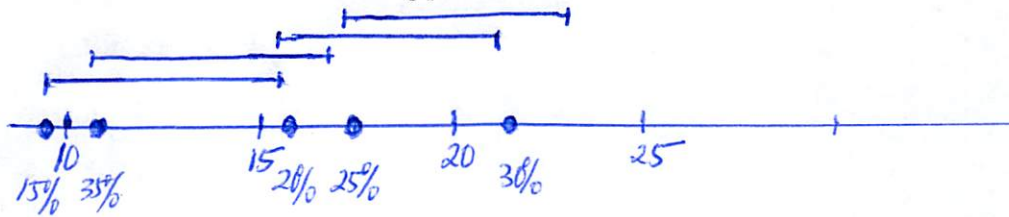
The Tukey multiple comparison method is more reliable. The Tukey "yardstick" for implementing the multiple comparison method is given by:

$$T_{\alpha} = q_{\alpha}(a, f) \sqrt{\frac{MSE}{n}},$$

where α is the preassigned significance level for the multiple comparison testing procedure, a is the number of means we are comparing and f is the df associated with the SSE, used to calculate the MSE. In our example, $a = 5$, $f = N - n = 20$, $n = 5$ and $MSE = 8.06$. The value of $q_{.05}(5, 20) = 4.24$, we obtain from Table V in the Appendix of our textbook on page 708. Therefore,

$$T_{.05} = 4.24 \sqrt{\frac{8.06}{5}} = 5.383.$$

As we can see the Tukey "yardstick" is larger than the LSD "yardstick". Based on this we examine the following plot:



Based on the above we can conclude that the mean breaking strength of the 30% blended fiber is not significantly different for the 25% blended fiber. The mean breaking strength of the 25% blended fiber is not significantly different for the 20% blended fiber. The mean breaking strength of the 25% blended fiber is not significantly different for the 35% blended fiber. Moreover, the mean breaking strength of the 35% blended fiber is not significantly different for the 15% blended fiber. Looking at the plot above we also note that the mean breaking strength of the 30% blended fiber is not connected by a line to the means of the 20%, 35% and the 15% blended fibers. One can make a weak recommendation that the 30% blended fiber appears to be the strongest from all but the 25% blended fiber.